

565. Array Nesting

You are given an integer array `nums` of length `n` where `nums` is a permutation of the numbers in the range $[0, n - 1]$.

You should build a set $s[k] = \{nums[k], nums[nums[k]], nums[nums[nums[k]]], \dots\}$ subjected to the following rule:

- The first element in $s[k]$ starts with the selection of the element `nums[k]` of index $= k$.
- The next element in $s[k]$ should be `nums[nums[k]]`, and then `nums[nums[nums[k]]]`, and so on.
- We stop adding right before a duplicate element occurs in $s[k]$.

Return the longest length of a set $s[k]$.

Example 1:

- **Input:** `nums = [5,4,0,3,1,6,2]`
- **Output:** 4
- **Explanation:**

`nums[0] = 5, nums[1] = 4, nums[2] = 0, nums[3] = 3, nums[4] = 1, nums[5] = 6, nums[6] = 2.`

One of the longest sets $s[k]$:

$$s[0] = \{nums[0], nums[nums[0]], nums[nums[nums[0]]], \dots\} = \{5, 6, 2, 0\}$$

Example 2:

- **Input:** `nums = [0,1,2]`
- **Output:** 1

Constraints:

- $1 \leq \text{nums.length} \leq 10^5$
- $0 \leq \text{nums}[i] < \text{nums.length}$
- All the values of `nums` are unique.